

Software Requirements Specification (SRS)



A Private University Social Network Platform

Prepared by — Team Mavericks:

Name	Student ID
Joydip Datta	0112330960
MD. Saem Ferdous	0112330950
Md. Monabbur Hosen Bhuiyan	0112330936
Md. Mahfujur Rahman Himel Akon	0112330935



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Contents

1	Introduction	3
1.1	Project Overview	3
1.2	Purpose of the System	3
1.3	Problem Statement / Motivation	3
2	Project Objectives	4
2.1	Social and Community Objectives	4
2.2	Professional and Career Development Objectives	4
2.3	Campus Utility and Technical Objectives	4
3	Target Users	5
3.1	Students	5
3.2	Alumni	5
3.3	Faculty	5
3.4	System Administrators	5
3.5	Transport Drivers	5
4	Benchmark Analysis	6
4.1	Competitor Overview	6
4.2	Feature Comparison Matrix	7
4.3	Key Differentiators	8
5	Gap Analysis	9
5.1	Identified Gaps in Existing Solutions	9
5.2	How UniConnecT Addresses the Gaps	10
6	Feasibility Analysis	12
6.1	Market Analysis	12
6.2	Cash Flow and NPV Analysis	14
7	MoSCoW Model	16
7.1	Must Have	16
7.2	Should Have	17
7.3	Could Have	17
7.4	Won't Have (This Release)	18
8	Kano Model	19
8.1	Basic (Must-Be) Quality Features	19
8.2	Performance Quality Features	20

8.3	Excitement (Delighter) Features	20
9	Functional Requirements	22
9.1	Authentication and Account Management	22
9.2	Social Feed and Content Management	22
9.3	Profiles, Connections, and Search	22
9.4	Jobs, Events, and Groups	23
9.5	Messaging, Notifications, and Mentorship	23
9.6	Campus Utilities, Administration, and Settings	24
10	Non-Functional Requirements	25
10.1	Usability, Performance, and Responsiveness	25
10.2	Access Control, Theming, and Maintainability	25
11	Pages / Modules	27
11.1	Public and Authentication Pages	27
11.2	Core Social and Professional Pages	27
11.3	Campus Utility Pages	28
11.4	Settings and Administration Pages	28
12	Tools and Technologies	29
12.1	Design and Prototyping Tools	29
12.2	Supporting and Auxiliary Tools	29
13	System Design / Wireframe	30
13.1	Authentication and Public Pages	30
13.2	Core Social and Professional Wireframes	32
13.3	Campus Utilities and Administration	35
13.4	System Architecture Diagrams	36
14	Usability / Design Considerations	38
14.1	Navigation Flow	38
14.2	Responsiveness Across Devices	38
14.3	Color Consistency and Design System	38
14.4	Accessibility Considerations	39
14.5	Key User-Friendly Design Decisions	39
15	Conclusion	41

1. Introduction

1.1 Project Overview

UniConnecT is a multi-tenant private social networking platform designed exclusively for university communities, connecting students, alumni, faculty, and administrative staff within a shared, institution-specific digital environment. Unlike general-purpose social networks, UniConnecT scopes all interactions, content, and connections to a single university per deployment, ensuring every engagement is relevant, verified, and community-driven. The platform serves as a central hub for academic and professional campus life, bringing together social engagement, career opportunities, mentorship, campus services, and real-time communication under one cohesive interface.

1.2 Purpose of the System

The purpose of UniConnecT is to eliminate the fragmentation that currently exists across university communication channels by providing one cohesive platform where every member of the university community can connect, collaborate, and grow. Through UniConnecT, students can build their professional network, discover job and internship opportunities, receive structured mentorship from alumni, stay informed through official university news, join groups and clubs, and coordinate campus activities — all within a verified, institution-bound environment. Faculty and administrators gain a structured channel to publish announcements, manage users and content, and engage with the broader campus community. Transport staff can broadcast live shuttle location data for student use.

1.3 Problem Statement / Motivation

University communities currently lack a dedicated digital space that serves their unique social and professional needs. Students rely on a fragmented combination of general social media platforms, university email chains, physical notice boards, and third-party job portals — none of which are tailored to the university context or integrated with one another. This fragmentation results in missed job and internship opportunities, weak alumni-to-student mentorship pipelines, poor information flow between administration and students, and no unified sense of campus identity online. Beyond social and professional concerns, practical campus services such as lost-and-found coordination and shuttle tracking are handled through informal channels with no digital structure. UniConnecT is proposed to address all of these challenges by offering a modern, purpose-built platform that brings the entire university experience — social, professional, academic, and logistical — into a single, well-designed digital space.

2. Project Objectives

The primary objectives of UniConnecT are grouped below by theme.

2.1 Social and Community Objectives

- **Unified Campus Social Feed:** Provide a community-driven social feed where students, alumni, and faculty can post, react, comment on, and share content relevant to their university.
- **Campus Events Management:** Enable creation, browsing, and RSVP for campus events so the community stays informed and engaged in university life.
- **Groups and Clubs:** Support the creation and management of university groups (clubs, departments, study circles) with dedicated feeds, file resources, study sessions, member roles, and group rules.
- **Real-Time Messaging:** Provide direct and group messaging for communication between connected community members and within mentorship relationships.

2.2 Professional and Career Development Objectives

- **Alumni–Student Mentorship:** Design a structured mentorship module enabling students to discover alumni mentors and request guidance on academic and career development.
- **Job and Internship Discovery:** Offer a university-internal job board where alumni and faculty can post opportunities and students can browse and apply, all within a trusted institutional network.
- **Professional Connection Network:** Implement a bidirectional LinkedIn-style connection system allowing members to build and grow a verified university professional network.

2.3 Campus Utility and Technical Objectives

- **Campus Utilities:** Integrate practical campus tools — a lost-and-found item tracking module and a real-time shuttle schedule and location tracker — directly into the platform.
- **Role-Based Access and Multi-Tenancy:** Design distinct, role-aware interface views for students, alumni, faculty, administrators, and transport drivers, with each university operating as a fully isolated tenant.

- **Modern, Accessible, Responsive Interface:** Deliver a visually consistent, accessible UI across desktop, tablet, and mobile screen sizes, built on a unified UIU-branded design system.

3. Target Users

The following user groups are the intended audience for UniConnecT. All users belong to a single university community; the platform does not allow cross-university interaction.

3.1 Students

The primary users of the platform. Students use UniConnecT to browse and post on the social feed, build connections with peers and alumni, discover jobs and internships, join groups and clubs, RSVP for events, request mentorship from alumni, access campus news, save drafts, report lost items, and track shuttle locations.

3.2 Alumni

Graduated members of the university who maintain access to the platform after leaving. Alumni can post job and internship opportunities, offer mentorship to current students, engage with the social feed, participate in groups, and stay professionally connected to the university community.

3.3 Faculty

Academic staff members who use the platform to publish announcements and news, engage with students and alumni, create and manage events, and participate in group discussions. Faculty may also act in a moderation capacity within their department-level groups.

3.4 System Administrators

University administrators who oversee the platform's operation. Administrators manage user accounts and roles, issue and revoke invitation tokens, moderate reported content, configure university-wide settings, and monitor platform activity through a dedicated admin panel. All administrative actions are recorded in an audit log.

3.5 Transport Drivers

A least-privilege role assigned exclusively to campus shuttle and transport staff. Drivers interact only with a dedicated GPS broadcast view to transmit live shuttle location data to the platform. They have no access to the broader social application.

4. Benchmark Analysis

Benchmark analysis evaluates UniConnecT against the five platforms that occupy the closest competitive space: purpose-built university alumni and community engagement SaaS products. Unlike general social networks, these platforms are specifically sold to higher-education institutions — making them the most relevant frame of reference for positioning UniConnecT’s value and identifying remaining market gaps.

4.1 Competitor Overview

- **PeopleGrove** (Direct) — A career-success and mentoring platform deployed at 500+ universities primarily in the USA. PeopleGrove excels at structured alumni-to-student career advising and job referral workflows. However, it does not offer a general social feed, real-time messaging, campus utilities, or university news. Pricing is custom and expensive, making it inaccessible to South Asian institutions operating on tighter budgets.
- **Graduway** (Direct) — An alumni community and engagement platform used by 2,750+ institutions globally. Graduway focuses on alumni directory management, mentorship matching, and fundraising pipelines. It is predominantly alumni-facing and does not serve current students, faculty, or transport staff as first-class actors. It lacks a social feed, real-time chat, and any campus utility layer. Enterprise pricing makes it prohibitive for emerging-market universities.
- **Almabase** (SaaS) — A fundraising and events management SaaS for alumni associations, priced at approximately \$3,000 per year. Almabase covers alumni directories, giving campaigns, and event ticketing but provides no student social networking, job board, mentorship economy, or campus utilities. Its feature surface is narrow relative to its cost.
- **AlumnForce** (SaaS) — A private social network for alumni associations, originating in Europe. AlumnForce offers alumni directories, a jobs board, and some social feed functionality, but the platform is alumni-only — current students and faculty are not integrated as participants. Real-time messaging is limited. Pricing is quote-based and typically enterprise-tier, with little published transparency.
- **Hivebrite** (SaaS) — A community-building and branding platform serving alumni networks, nonprofits, and professional associations. Hivebrite offers events, a directory, a jobs board, and a community feed, but it is not university-specific — any organisation can adopt it. It carries no campus utility features and no institutional governance layer tailored to the university context. Pricing is enterprise-only with no self-serve tier.

4.2 Feature Comparison Matrix

Feature	People-Grove	Gradu-way	Alma-base	Alumn-Force	Hive-brite	Uni-ConnecT
Students as first-class users	Partial	No	No	No	No	Yes
Faculty & admin roles	No	No	No	No	No	Yes
University invite verification	Yes	Yes	Yes	Yes	Partial	Yes
Social feed (posts/reactions)	No	Partial	No	Partial	Yes	Yes
Real-time messaging	Partial	No	No	Partial	Partial	Yes
Structured mentorship flow	Yes	Yes	Partial	Partial	Partial	Yes
Points / reward economy	No	No	No	No	No	Yes
Integrated job board	Yes	Partial	Partial	Yes	Yes	Yes
Campus events management	No	Yes	Yes	Yes	Yes	Yes
University news & content sync	No	No	No	No	No	Yes
Lost & found module	No	No	No	No	No	Yes
Live shuttle GPS tracking	No	No	No	No	No	Yes
Multi-tenant SaaS architecture	Yes	Yes	Yes	Yes	Yes	Yes
Role-based audit log	No	No	No	No	No	Yes
Affordable / transparent pricing	No	No	Partial	No	No	Yes

Table 1: Feature comparison of UniConnecT against direct university-community competitors.

4.3 Key Differentiators

The matrix reveals that existing platforms cluster into two problem spaces: alumni engagement tools (Graduway, Almabase, AlumnForce) and generic community platforms (Hivebrite) — with PeopleGrove the closest competitor on the mentorship dimension. UniConnecT holds an exclusive position across four dimensions that no single competitor addresses:

- **Whole-community coverage:** UniConnecT is the only platform where students, alumni, faculty, administrators, and transport drivers all participate as first-class, role-differentiated actors. Every competitor targets alumni as the primary user and treats students as secondary or absent.
- **Integrated mentorship economy:** While PeopleGrove and Graduway offer mentorship matching, none implement a points-based reward economy with gift card redemption, session logging, or alumni capacity enforcement. UniConnecT creates mutual incentive on both sides of the mentorship relationship.
- **Campus utility layer:** Lost-and-found management and live shuttle GPS tracking (driver broadcast + client-side route interpolation) are absent from every benchmarked platform. These features make UniConnecT indispensable for day-to-day campus life, not just career development.
- **South Asia pricing accessibility:** All five competitors use enterprise or custom pricing models opaque to budget-constrained South Asian universities. UniConnecT's transparent, affordable subscription (\$500/month) is specifically designed for the 54+ private universities in Bangladesh and comparable institutions across the region.

5. Gap Analysis

Gap analysis identifies the unmet needs within the existing landscape of tools used by university communities and maps each gap to a concrete UniConnecT feature.

5.1 Identified Gaps in Existing Solutions

1. **No verified, university-bound social space.** Students and alumni use general platforms (Instagram, Facebook) for campus discourse. These are public, unverified, and carry no institutional accountability. Misinformation, spam, and off-topic content are endemic.
2. **Weak alumni-to-student mentorship pipelines.** Career services departments at most universities manage mentorship through spreadsheets and email. There is no structured platform to match mentees with mentors, track session progress, or reward alumni participation.
3. **Fragmented professional discovery.** Job and internship postings are scattered across notice boards, department WhatsApp groups, and external portals. No platform consolidates alumni-posted opportunities scoped to the university.
4. **Undigitised campus utility services.** Lost-and-found is managed via physical boards or informal group chats. Shuttle schedules are static PDFs. Live vehicle tracking, item reporting, and claim workflows have no digital home.
5. **No unified campus news and announcements.** Faculty announcements, club events, and administrative notices travel through multiple disconnected channels (email, Facebook, WhatsApp). Students miss critical information.
6. **Absent governance and audit trail.** General platforms offer no institutional moderation layer. Administrators cannot manage roles, issue verified invitation tokens, or maintain a tamper-evident log of platform actions.

5.2 How UniConnecT Addresses the Gaps

#	Gap	UniConnecT Resolution
1	Unverified social space	Invite-token registration + <code>university_id</code> -scoped data; all content is institution-bound.
2	No mentorship pipeline	Dedicated mentorship module with request/accept flow, session logs, points economy, and alumni capacity control.
3	Fragmented job discovery	Alumni-authored job board with role-gated posting, application tracking, and university-scoped visibility.
4	Undigitised campus utilities	Lost-&-found CRUD module + shuttle module with driver GPS broadcast and client-side route interpolation.
5	Siloed news and announcements	Unified news module fed by admin-authored posts and auto-imported WordPress/external content via the content-sync engine.
6	No governance or audit trail	Admin panel with invitation management, role assignment, content reporting, and <code>university_audit_log</code> .

Table 2: Mapping of identified gaps to UniConnecT features.

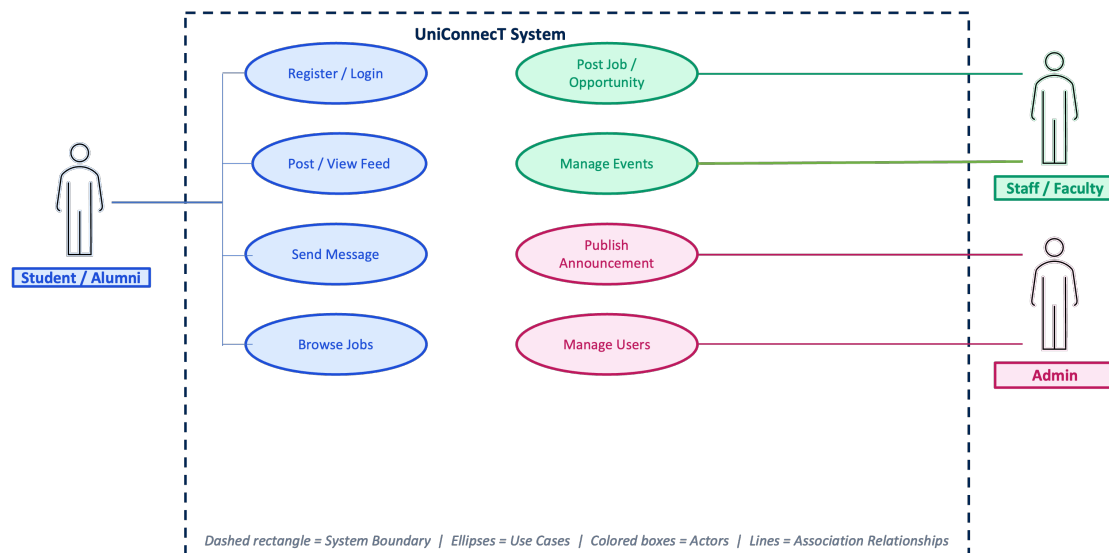


Figure 1: Use-case diagram for UniConnecT. Three actor groups — Student/Alumni, Staff/Faculty, and Admin — interact with core use cases including Register/Login, Post/View Feed, Send Message, Browse Jobs, Post Job/Opportunity, Manage Events, Publish Announcement, and Manage Users. Lines indicate association relationships between actors and the use cases they are permitted to invoke within the system boundary.

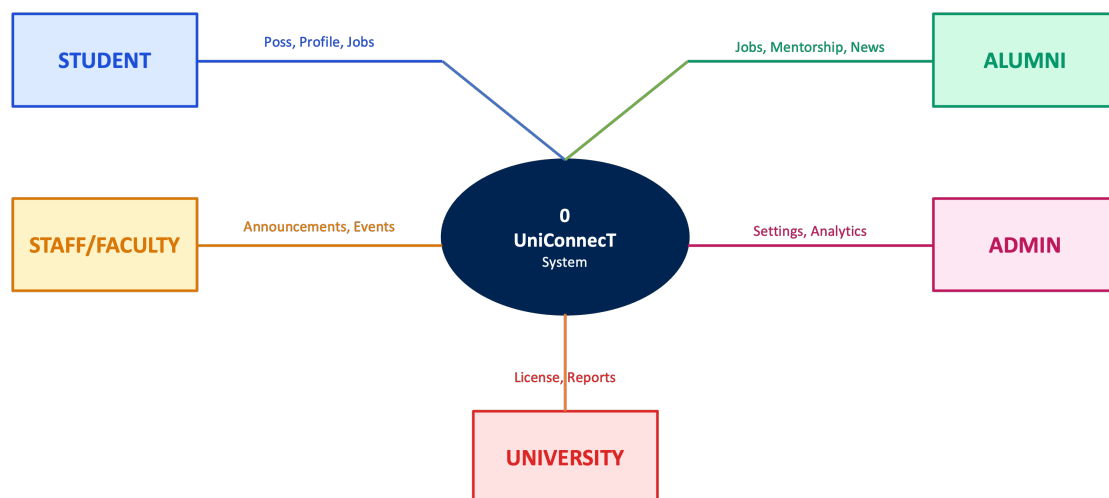


Figure 2: System context diagram (Level 0 DFD). UniConnecT occupies the central node. Five external entities interact with the system: Student (Posts, Profile, Jobs), Alumni (Jobs, Mentorship, News), Staff/Faculty (Announcements, Events), Admin (Settings, Analytics), and University (License, Reports). Data flows on each edge represent the primary information exchanged between each entity and the platform.

6. Feasibility Analysis

Feasibility analysis examines whether UniConnecT is viable from market, technical, and financial perspectives within the two-year horizon of the project.

6.1 Market Analysis

Total addressable market. Bangladesh has 54 approved private universities (UGC data, 2025) with a combined active student enrolment exceeding 600,000. Each university operates at least one administrative domain and runs independent communication channels. There is currently no university-specific SaaS social platform serving this market.

Serviceable addressable market. Of the 54 private universities, approximately 30 have active digital presences, dedicated IT departments, and the budget infrastructure compatible with a SaaS subscription model. The serviceable market is therefore ≈ 30 universities in the first two years.

Serviceable obtainable market (2-year target). A conservative penetration rate of 33% of the serviceable market yields a two-year target of 10 university deployments. At a per-university subscription of \$500 per month, the obtainable annual recurring revenue at full Year 2 capacity is \$60,000 USD.

Pricing rationale. The subscription covers hosting (Azure App Service + Neon PostgreSQL + Redis), cloud storage (Cloudflare R2), email delivery (Resend), and ongoing maintenance. At \$500/month, the break-even unit cost per deployment is below \$200/month at 10+ active tenants, yielding a healthy gross margin above 60%.

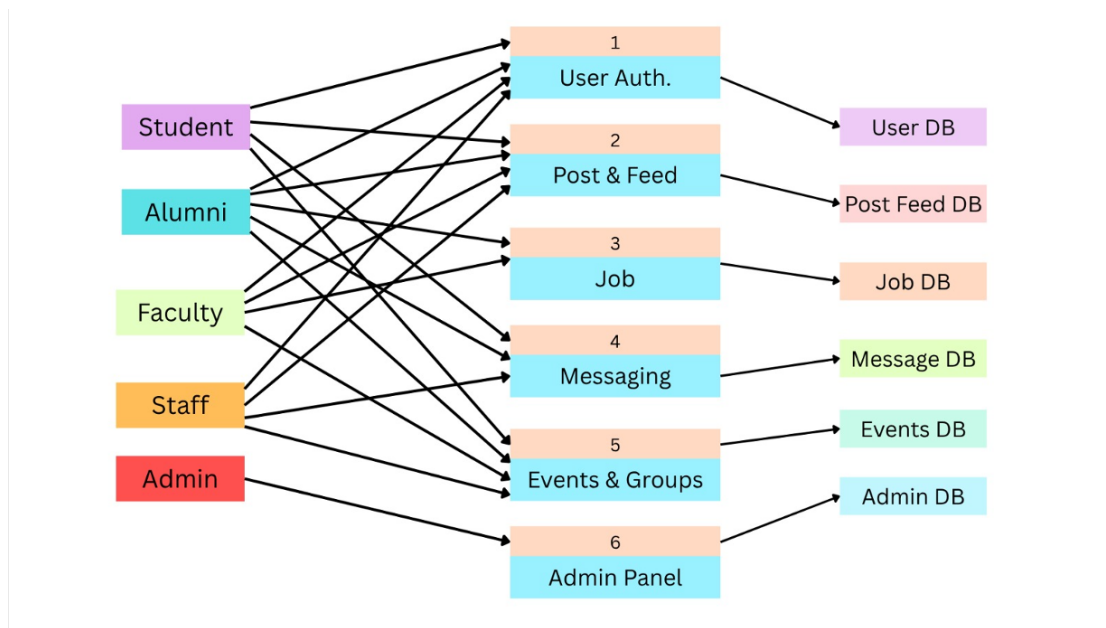


Figure 3: Data flow diagram (Level 1) for UniConnecT. Shows how data originates from five actor types (Student, Alumni, Faculty, Staff, Admin), flows through six processing modules (User Auth, Post & Feed, Job, Messaging, Events & Groups, Admin Panel), and is persisted in six data stores (User DB, Post Feed DB, Job DB, Message DB, Events DB, Admin DB). Arrows indicate the direction of data movement between actors, processes, and stores.

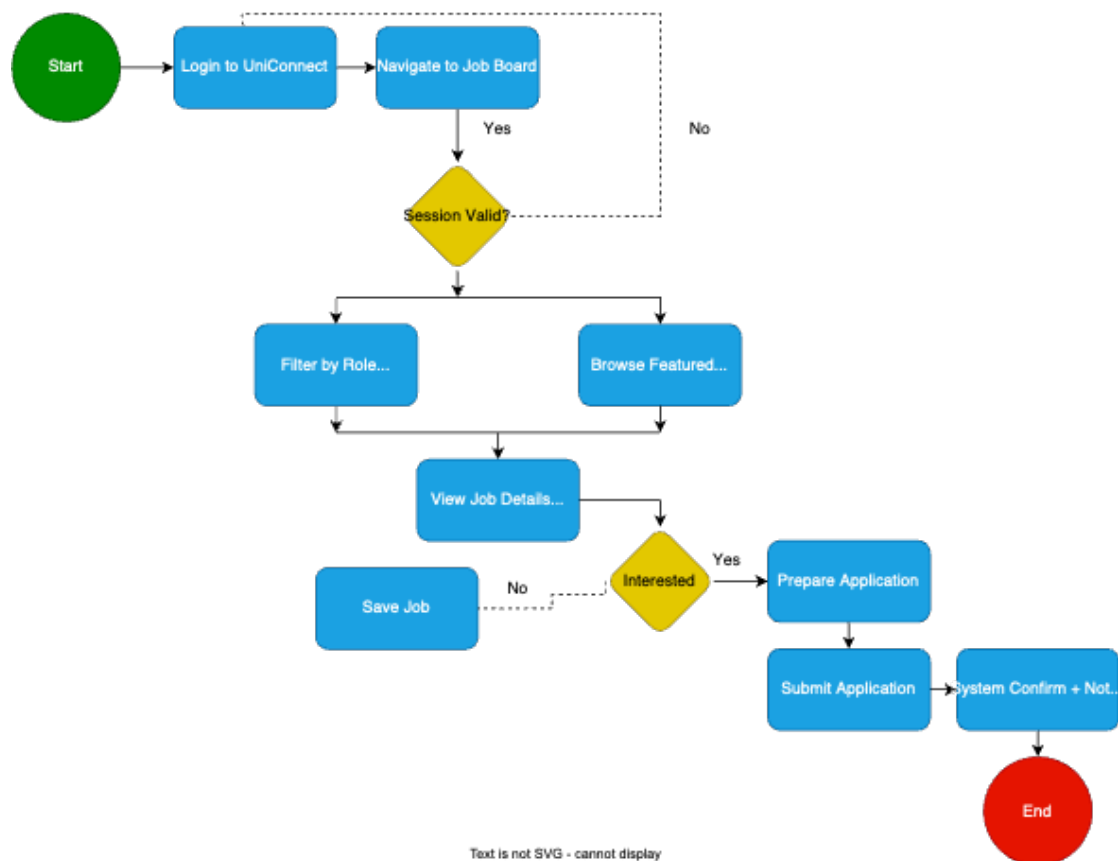


Figure 4: Activity diagram illustrating a student’s end-to-end job application journey on UniConnecT. The flow begins at Login, proceeds to the Job Board (gated by a Session Valid decision), allows the user to Filter by Role or Browse Featured listings, view job details, and then branches on interest: saving the job or preparing and submitting an application, which concludes with a system confirmation and notification.

6.2 Cash Flow and NPV Analysis

The financial model below projects costs and revenues over a 24-month window (July 2026 — June 2028) using a discount rate of **10% per annum**, which approximates the opportunity cost of capital for an early-stage software project.

Assumptions:

- Development team: 4 members × 6 months at \$2,000/month each = \$48,000 total initial investment (Year 0, already incurred in development).
- Operating costs Year 1: hosting \$3,600 + marketing \$2,400 + maintenance support \$6,000 = \$12,000.
- Operating costs Year 2: hosting \$7,200 + marketing \$4,800 + maintenance support \$8,000 = \$20,000.

- Universities onboarded: 4 in Year 1, scaling to 12 by end of Year 2.
- Average subscription: \$500/month per university.

Item	Year 0 (\$USD)	Year 1 (\$USD)	Year 2 (\$USD)
<i>Inflows</i>			
Subscription revenue	—	24,000	72,000
<i>Outflows</i>			
Initial development cost	(48,000)	—	—
Operating costs	—	(12,000)	(20,000)
Net cash flow	(48,000)	12,000	52,000
Discount factor $(1.10)^n$	1.0000	1.1000	1.2100
Present value of net cash flow	(48,000.00)	10,909.09	42,975.21
Cumulative NPV	(48,000.00)	(37,090.91)	+5,884.30

Table 3: Two-year cash flow projection and NPV calculation (discount rate 10%).

Net Present Value (NPV):

$$\text{NPV} = -48,000 + \frac{12,000}{(1.10)^1} + \frac{52,000}{(1.10)^2} = -48,000 + 10,909.09 + 42,975.21 = \mathbf{\$5,884.30}$$

Since $\text{NPV} > 0$, the project generates a positive return within the two-year window at the assumed 10% discount rate. Payback is achieved during Year 2 as cumulative discounted cash flows cross break-even at approximately Month 21.

Sensitivity note. The NPV is sensitive to university acquisition rate. Signing just 2 additional universities in Year 1 (6 total instead of 4) raises Year 1 revenue by \$12,000 and improves NPV to approximately \$16,800 — a 185% increase in project value, underscoring the leverage of early client acquisition.

7. MoSCoW Model

The MoSCoW model categorises all UniConnecT features by priority to guide development sequencing. The model reflects the current state of the implemented codebase and was validated against stakeholder requirements.

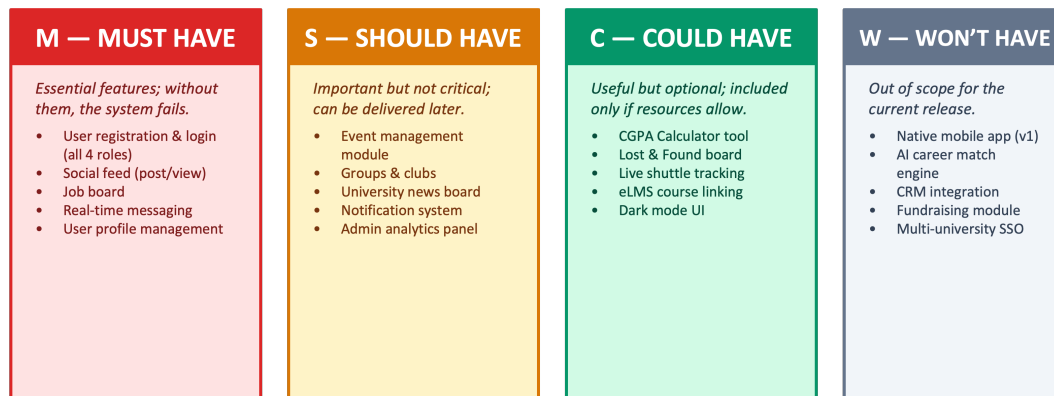


Figure 5: MoSCoW prioritisation matrix for UniConnecT. The four quadrants (Must Have, Should Have, Could Have, Won't Have) map each feature to its delivery commitment for the current release. Features in the Must Have quadrant were treated as hard scope; their absence would render the platform non-functional for the target university community.

7.1 Must Have

These features are non-negotiable and are fully implemented in the current release. Without them, the platform cannot operate as a university social network.

- **Invite-token registration with email OTP verification** — ensures only verified university members can join.
- **JWT authentication with silent refresh** — stateless, secure session management with 15-minute access tokens and 7-day refresh cookies.
- **Role-based access control** (student, alumni, faculty, admin, driver) — each role sees and can perform only what the university authorises.
- **Social feed** (create, react, comment, share posts) — the core engagement surface of the platform.
- **Real-time messaging** (direct, group, mentorship threads) — Socket.io powered, scaled via Redis pub/sub.
- **User profiles with experience, education, and skills** — foundation of professional identity on the platform.

- **University-scoped data isolation** — multi-tenant architecture; every DB row carries `university_id`; no cross-university data leakage.
- **Admin panel** (user management, invitations, content reports) — universities cannot operate the platform without institutional governance.
- **Notifications** (in-app + Web Push) — users must be informed of relevant activity in real time.

7.2 Should Have

These features significantly enhance the value proposition and are implemented in the current release, though the platform would still function at a basic level without them.

- **Job and internship board** — alumni-posted opportunities scoped to the university; application tracking for students.
- **Events management** with RSVP — creation, browsing, and attendance tracking for campus events.
- **Groups and clubs** — private/public groups with member roles, resources, study sessions, pinned posts, and group rules.
- **Mentorship module** — request/accept flow, session logging, points economy, alumni capacity enforcement, gift card redemption.
- **Full-text search** — Postgres generated `search_vector` columns across profiles, posts, jobs, events, and groups.
- **University news feed** with content-sync engine — auto-import from WordPress/external CMS with admin review workflow.
- **Connections (LinkedIn-style)** — bidirectional connection graph replacing the old follow model.
- **Profile analytics** (7/30/90-day view counts, viewer list) — professional insight into profile reach.

7.3 Could Have

These features add value but were treated as stretch goals. A subset is implemented; the remainder is scoped for future iterations.

- **Draft posts across content types** — save-and-resume authoring for posts, jobs, events, and news articles. (*Implemented.*)

- **Explore page** with trending tags and suggested users. *(Implemented.)*
- **Resume export** from profile data to a downloadable PDF. *(Implemented.)*
- **Dark/light theme toggle** with user preference persistence. *(Implemented.)*
- **Presence indicators** — real-time online/offline status with privacy tier controls. *(Implemented.)*
- **Group digest email** (weekly summary of group activity via Bull cron). *(Implemented.)*
- **Mobile-responsive design** — full Tailwind CSS responsive layout. *(Implemented.)*
- **Skyvern browser-automation fallback** for content sync when WordPress REST API is unavailable. *(Implemented.)*

7.4 Won't Have (This Release)

These features were explicitly deferred to avoid scope creep. They remain on the long-term product roadmap.

- **Native mobile applications** (iOS / Android) — the current release is a progressive web app only; native apps require a separate development cycle.
- **Video calling or live streaming** — real-time video would require WebRTC infrastructure and per-stream bandwidth cost modelling beyond current scope.
- **AI-powered feed curation** — machine learning recommendations require training data volume only achievable after several months of live usage.
- **Cross-university alumni network** — multi-university interaction conflicts with the institution-bound privacy model and requires a governance redesign.
- **Integrated payment gateway** — monetised mentorship sessions and paid events are deferred pending legal and financial compliance review.
- **User self-deletion** — account deletion is admin-mediated (request-with-reason workflow) to preserve audit integrity and data retention compliance.

8. Kano Model

The Kano model classifies UniConnecT features by the nature of their relationship to user satisfaction. It distinguishes between features that users expect as table-stakes (Basic), features where more is always better (Performance), and features that unexpectedly delight users when present (Excitement).

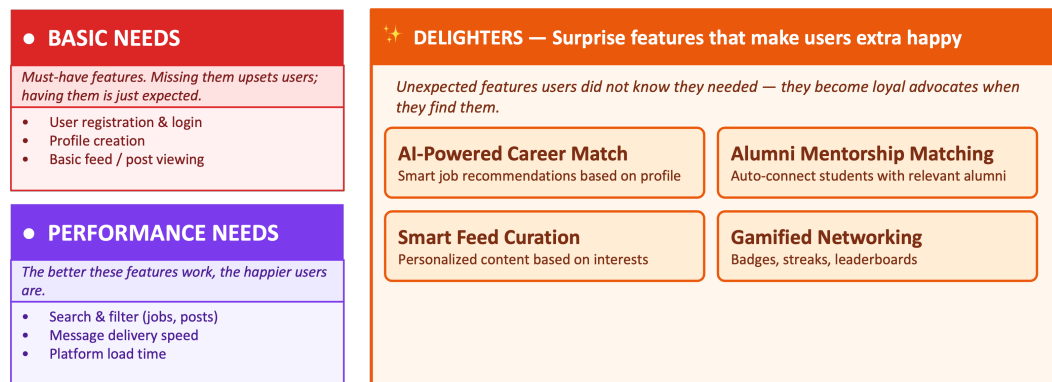


Figure 6: Kano model overview for UniConnecT, organised as three feature categories. *Basic Needs* (must-be quality): user registration/login, profile creation, and basic feed viewing — their absence causes immediate dissatisfaction. *Performance Needs*: search & filter, message delivery speed, and platform load time — satisfaction scales linearly with quality. *Delighters*: AI-powered career match, alumni mentorship matching, smart feed curation, and gamified networking — unexpected features that create strong positive reactions when discovered.

8.1 Basic (Must-Be) Quality Features

These features are taken for granted when present; their absence causes immediate dissatisfaction and platform rejection. They are threshold requirements, not differentiators.

- **Secure login and session management** — users expect authentication to simply work. Any failure here (token leaks, silent logouts) destroys trust immediately.
- **Fast page loads and responsive UI** — users expect sub-2-second render times on any device. Slowness is a hygiene failure.
- **Reliable notification delivery** — missing a job offer notification or mentorship acceptance is an unacceptable failure, not a missing feature.
- **Data privacy and access isolation** — users expect their content to be visible only to their university community. A cross-tenant data leak would terminate the product.

- **Mobile-readable layout** — the majority of students access platforms on mobile. A desktop-only layout would exclude a large fraction of the user base.
- **Profile completeness** — name, photo, role, and department are expected to be present and editable. An incomplete identity layer undermines trust in every other interaction.

8.2 Performance Quality Features

These features directly scale satisfaction with the quality of their implementation. The better the feature works, the happier the user; degraded performance causes proportional dissatisfaction.

- **Feed relevance and ranking** — a chronological feed is baseline; the HN-style `hot_score` ranking algorithm and “Top” feed sort increase satisfaction linearly as relevance improves.
- **Search accuracy and speed** — the quality of full-text search (Postgres GIN-indexed `search_vector` with tsvector weighting A/B/C) maps directly to how quickly users find peers, jobs, or content.
- **Mentorship match quality** — the usefulness of the mentorship module scales with how well students can discover the right mentor by field, seniority, and availability.
- **Job board coverage** — the more alumni post opportunities, the more valuable the board. Coverage is the primary performance dimension of this feature.
- **Message delivery latency** — real-time chat satisfaction correlates directly with message round-trip latency. Sub-100ms delivery (Redis pub/sub, local Socket.io) feels instant; anything above 500ms feels broken.
- **Content sync freshness** — the news module’s value scales with how recently the WordPress importer has run. Stale news is worse than no news.

8.3 Excitement (Delighter) Features

These features are not anticipated by users. When discovered, they generate strong positive reactions and become key advocates of the platform. Their absence causes no dissatisfaction.

- **Points-based mentorship economy with gift card redemption** — alumni earn redeemable points for mentorship sessions. No university platform offers this; users who discover it are strongly motivated to participate.

- **Live shuttle GPS tracking** — students discovering real-time bus location on a campus social app is wholly unexpected and immediately useful.
- **Resume PDF export from profile** — turning the platform profile into a downloadable resume surprises users accustomed to maintaining separate CVs.
- **Skyvern-powered news auto-import** — administrators discover that the platform automatically harvests university news from external CMS sources via browser automation when the standard API is unavailable.
- **Profile view analytics** (7/30/90-day viewer list) — users do not expect to see who has viewed their profile. Discovering this feature creates immediate engagement loops and motivates profile completion.
- **Presigned S3 upload flow** — file attachments upload directly from the browser to cloud storage with zero server-side byte handling. Users experience this as surprising speed; developers appreciate the architectural elegance.

9. Functional Requirements

The functional requirements define the specific behaviors and capabilities the UniConnecT system shall provide. Each requirement is labeled with a unique identifier for traceability.

9.1 Authentication and Account Management

- FR-1:** The system shall allow new users to register using a valid university invitation token, providing their full name, email address, and a chosen password.
- FR-2:** The system shall send a one-time password (OTP) to the user's registered email address upon registration, which must be confirmed to activate the account.
- FR-3:** The system shall allow registered users to log in with their email and password and log out securely from any session.
- FR-4:** The system shall support a password recovery flow that allows users to request a reset link via their registered university email address.
- FR-5:** The system shall validate all user input during registration and login, displaying clear inline error messages for missing or invalid fields.

9.2 Social Feed and Content Management

- FR-6:** The system shall display a personalized social feed upon login, showing posts, reactions, and activity from the user's university community, sortable by "Recent" and "Top" (engagement-ranked) modes.
- FR-7:** The system shall allow users to create, edit, delete, and save draft posts on the social feed, with support for text content, images, and links.
- FR-8:** The system shall allow users to react to and comment on any post in the feed, with real-time count updates.

9.3 Profiles, Connections, and Search

- FR-9:** The system shall allow users to view and edit their own profile, including profile photo, bio, work experience entries, education history, skills, and featured items (maximum five).
- FR-10:** The system shall track profile views and provide each profile owner with an analytics view showing visitor counts over the past 7, 30, and 90 days, along with a list of recent viewers.

FR-11: The system shall allow users to send, accept, decline, and withdraw connection requests. Accepted connections form a persistent, bidirectional professional network visible on each user's profile.

FR-12: The system shall display connection suggestions to users based on mutual connections and shared university affiliation.

FR-13: The system shall provide a unified full-text search interface through which users can search for people, posts, jobs, events, groups, and news by keyword, with filterable result tabs per content type.

9.4 Jobs, Events, and Groups

FR-14: The system shall allow alumni and faculty to post job and internship listings specifying role title, description, requirements, and application deadline.

FR-15: The system shall display job listings in a filterable, sortable list view, with a detail page for each listing.

FR-16: The system shall allow users to create and browse campus events, with RSVP functionality and a detail view showing date, location, description, and confirmed attendee count.

FR-17: The system shall allow users to create, join, and leave groups (clubs, departments, study circles), with public and private group types. Private groups require a join request approved by a group administrator.

FR-18: The system shall allow group administrators to manage member roles (admin, moderator, member), pin posts, post group rules, share file resources with view tracking, and schedule study sessions with RSVP.

9.5 Messaging, Notifications, and Mentorship

FR-19: The system shall provide real-time direct messaging between any two connected users, and group messaging within shared conversation threads.

FR-20: The system shall automatically create a dedicated conversation thread for each accepted mentorship request, linking the student and the alumni mentor within the messaging module.

FR-21: The system shall deliver real-time in-app notifications for events including new connection requests, accepted connections, new messages, post reactions, mentions, event updates, and mentorship activity.

FR-22: The system shall allow students to send mentorship requests to alumni, specifying their goals and area of guidance sought. Alumni may accept or decline requests subject to their configured maximum mentee capacity.

9.6 Campus Utilities, Administration, and Settings

FR-23: The system shall allow users to browse and read official university news and announcements published by faculty or administrators, with a detail page per article.

FR-24: The system shall provide a lost-and-found module where users can post reports of lost or found campus items, including a description, location, and optional photo.

FR-25: The system shall display campus shuttle route schedules and real-time shuttle location, updated continuously from driver GPS broadcasts.

FR-26: The system shall allow drivers (driver role exclusively) to broadcast their GPS location via a dedicated interface, which is reflected live on the student-facing shuttle tracker.

FR-27: The system shall provide an administrator panel through which admins can manage user accounts, assign and change roles, create and revoke invitation tokens (including bulk invitations), resolve content reports, and configure university-wide settings.

FR-28: The system shall aggregate all of a user's unpublished drafts — across posts, job listings, news articles, and events — into a single drafts view accessible only by the author.

FR-29: The system shall allow users to configure notification preferences, privacy settings (profile visibility, presence visibility, message accessibility), and application appearance (light or dark theme) from a dedicated settings hub.

10. Non-Functional Requirements

Non-functional requirements specify the quality attributes and operational constraints the UniConnecT design shall satisfy.

10.1 Usability, Performance, and Responsiveness

NFR-1: Usability: The interface shall be intuitive and learnable, enabling a new student user to complete primary tasks — registration, OTP verification, profile setup, browsing the feed, and sending a connection request — without external documentation or guidance.

NFR-2: Consistency: All pages shall adhere to the UniConnecT design system, maintaining a consistent color palette, typography (font weight restricted to 400 and 500), spacing scale, border style (0.5px structural borders), and component visual language throughout. No hardcoded color values shall appear outside the defined token set.

NFR-3: Responsiveness: The interface shall render correctly and remain fully usable across three primary breakpoints: mobile (360px–767px), tablet (768px–1023px), and desktop (1024px and above). Navigation shall collapse to a bottom tab bar on mobile; multi-column layouts shall reflow to single-column views.

NFR-4: Performance (Perceived): The feed, dashboard, profile, and listing pages shall use skeleton loading states during data load, ensuring the interface appears immediately responsive and never presents a blank screen to the user.

NFR-5: Accessibility: All text-on-background color combinations in both the dark and light themes shall meet a minimum 4.5:1 contrast ratio (WCAG 2.1 Level AA). All interactive elements shall carry visible focus states. All form inputs shall have descriptive labels; icon-only controls shall include tooltips.

NFR-6: Feedback and Error Handling: The design shall include explicit visual states for every user action: button hover, loading, success, and error. Inline validation errors shall appear adjacent to the failing field. All list and feed views shall have a designed empty state rather than a blank area.

10.2 Access Control, Theming, and Maintainability

NFR-7: Role-Based Interface Separation: The design shall strictly enforce role-aware navigation and page access. The admin panel shall not surface in navigation for non-admin roles. The driver GPS broadcast view shall be accessible

only to driver-role users. Students, alumni, and faculty shall each see a navigation set scoped to their role's permitted features.

NFR-8: Dual-Theme Support: The design system shall define both a *Warm Futuristic Dark* default theme (navy surfaces, UIU orange accent, indigo interactive states) and a *Warm Neutral Light* theme. All components shall be designed to render correctly in both themes without design-level modifications per page.

NFR-9: Modifiability: The Figma design system shall use shared component libraries (atoms and molecules) so that any design system change — color, typography, spacing, border radius — propagates consistently across all pages without requiring individual page edits.

NFR-10: Readability: Body text shall be set at a minimum of 14px with adequate line-height. Section headings shall establish hierarchy through size and weight differences alone, not color. All UI labels and button text shall use sentence case; no ALL CAPS labels shall appear anywhere in the interface.

11. Pages / Modules

The following pages and modules constitute the UniConnecT Figma prototype. Each entry represents a distinct screen or logical section of the planned application.

11.1 Public and Authentication Pages

1. **Landing Page** — The public entry point of UniConnecT, presenting the platform's value proposition with a hero section and calls to action ("Join Your University Network", "Log In").
2. **Registration Page** — An invitation-token-gated sign-up form collecting the user's full name, email address, and password.
3. **OTP Verification Page** — A code-entry screen for confirming the user's email address using the one-time password sent upon registration.
4. **Login Page** — A secure sign-in form for registered users with inline validation and a "Forgot Password" link.
5. **Forgot Password / Reset Password Page** — A two-step recovery flow: the user submits their email, then follows a link to set a new password.

11.2 Core Social and Professional Pages

6. **Social Feed (Home / Dashboard)** — The primary authenticated screen displaying a ranked, scrollable feed of community posts, with "Recent" and "Top" sort tabs, a left-side profile summary panel, and a right-side discovery sidebar.
7. **Post Detail Page** — A full-page view of a single post showing the complete content, all reactions, and a threaded comment section.
8. **Profile Page** — Displays a user's public profile: cover photo, avatar, bio, experience, education, skills, featured items, recent activity, and profile analytics. Includes an edit flow for the profile owner.
9. **Connections / My Network Page** — Shows the user's accepted connections, incoming pending requests, sent requests, and a "People You May Know" suggestions section.
10. **Explore Page** — A discovery screen for finding new community members, trending content, and activity outside the user's immediate connection network.
11. **Search Results Page** — A unified results screen with filterable tabs for People, Posts, Jobs, Events, Groups, and News.

12. **Jobs Page** — A filterable, sortable listing of job and internship postings within the university network, with a detail page per listing and a posting form for eligible users.
13. **Events Page** — A listing of upcoming campus events with RSVP functionality and a full detail view per event.
14. **Groups Page** — A directory of university groups and clubs, with a detail view per group showing the group feed, members list, file resources, scheduled study sessions, pinned posts, and group rules.
15. **Messages Page** — A real-time messaging interface with a conversation list sidebar and an active chat panel, supporting direct and group conversations as well as dedicated mentorship threads.
16. **Notifications Page** — A chronological list of all in-app notifications for the logged-in user, covering connections, messages, reactions, mentions, and mentorship activity.
17. **News Page** — A listing of official university news and announcements, with a detail view per article including title, body, and publication date.
18. **Mentorship Page** — A module where students browse alumni mentor profiles and submit mentorship requests, and where alumni manage their active mentees, session records, and mentee capacity.

11.3 Campus Utility Pages

19. **Lost & Found Page** — A campus utility module for posting and browsing reports of lost or found items, with descriptions, last-known location, and optional photo.
20. **Shuttle Tracker Page** — Displays campus shuttle route schedules and a live shuttle position map updated from driver GPS broadcasts.
21. **Driver GPS Broadcast View** — A minimal, role-restricted interface for transport drivers to broadcast their current GPS location to the platform.

11.4 Settings and Administration Pages

22. **Drafts Page** — A personal page aggregating all of the user's unpublished drafts across posts, job listings, news articles, and events.
23. **Settings Page** — A multi-section hub covering appearance (light / dark theme), account management (deactivation, username), notification preferences, and privacy controls (profile, presence, and message visibility).

24. **Admin Panel** — A role-restricted management interface for administrators covering user management, role assignment, invitation management (individual and bulk), content report resolution, platform statistics, and university configuration.

12. Tools and Technologies

This section covers the design and prototyping tools used for UniConnecT. It describes the design process only.

12.1 Design and Prototyping Tools

- **Figma** — Primary tool for all UI/UX work: wireframing, high-fidelity mockups, interactive prototyping, and the shared component design system. All 24 pages and user flows are designed within Figma.
- **Figma Plugins:**
 - *Iconify* — Used to source and insert scalable vector icons (Phosphor Icons and Heroicons sets) directly into Figma frames.
 - *Unsplash* — Used to insert high-quality placeholder photography for profile avatars, event covers, and news thumbnails within mockups.
 - *A11y — Color Contrast Checker* — Used to verify that all text and UI element color pairings meet WCAG 2.1 Level AA contrast requirements across both dark and light themes.
- **FigJam** — Used to create the navigation flow diagram and module architecture diagram, leveraging FigJam’s drag-and-drop connector tools and sticky-note layout for clear, readable system diagrams.

12.2 Supporting and Auxiliary Tools

- **Canva** — Used for supplementary graphics including the project cover slide, presentation visuals, and promotional design assets.
- **Google Fonts** — Typography selections (Inter for UI labels and body text) sourced from Google Fonts and applied within the Figma design system.
- **Coolers** — Used to explore and finalize the UniConnecT color palette for both the Warm Futuristic Dark theme (navy, UIU orange, indigo) and the Warm Neutral Light theme.

13. System Design / Wireframe

All visuals in this section are taken from the Figma design prototype. The figures below present wireframes and UI sketches for the major pages of UniConnecT, along with a navigation flow diagram and module architecture diagram.

13.1 Authentication and Public Pages

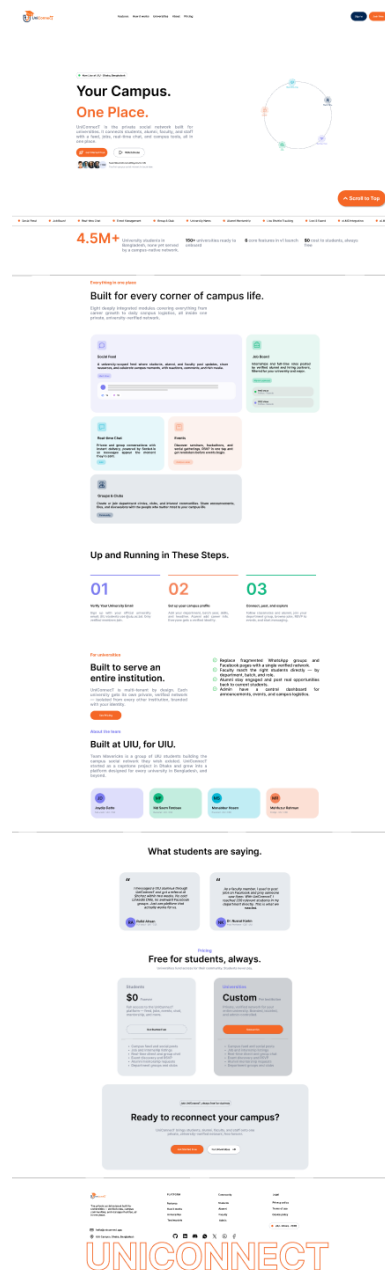


Figure 7: Figma Wireframe — Landing Page. The public entry screen with a hero section presenting UniConnecT’s value proposition, a top navigation bar, and calls to action for new and returning users.

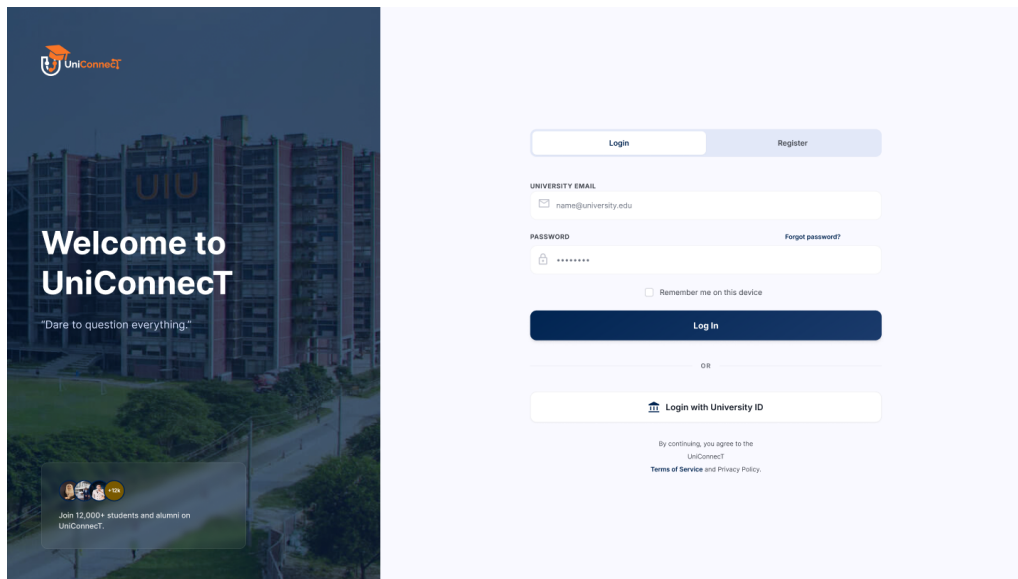


Figure 8: Figma Wireframe — Login Page. Email and password input form with inline validation states, a “Forgot Password” link, and the university domain context displayed in the header.

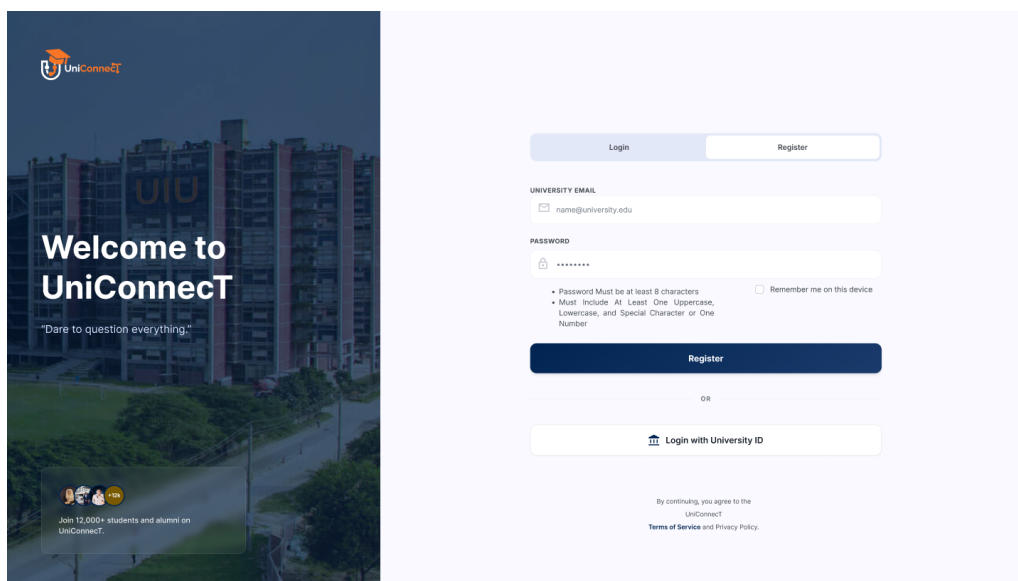


Figure 9: Figma Wireframe — Registration Page. Invitation-token-gated sign-up form collecting name, email, and password, followed by the OTP email verification step.

13.2 Core Social and Professional Wireframes

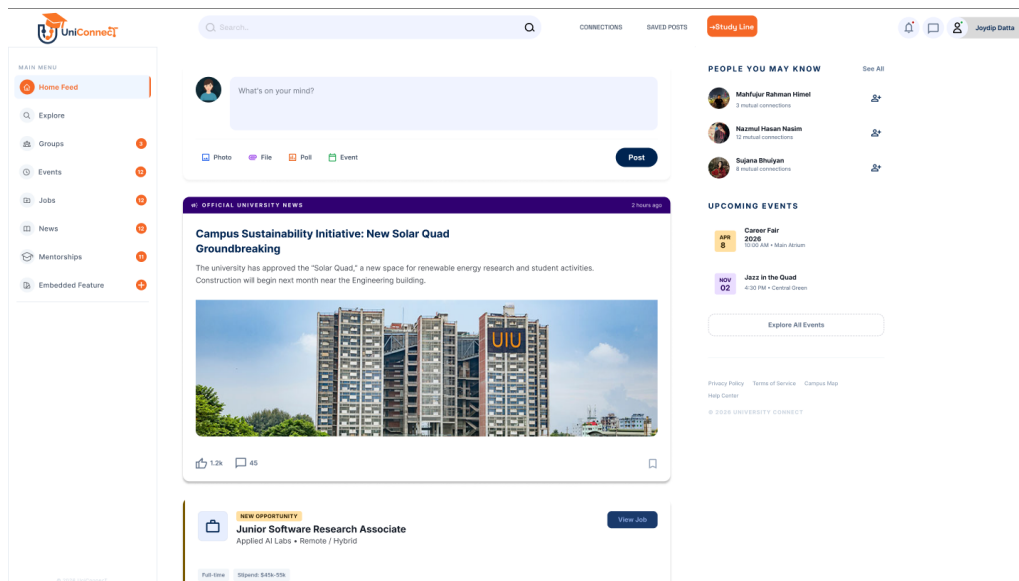


Figure 10: Figma Wireframe — Social Feed (Dashboard). The main authenticated screen: ranked post feed in the centre, a profile summary panel on the left, and a discovery/suggestions sidebar on the right. “Recent” and “Top” feed tabs are shown.

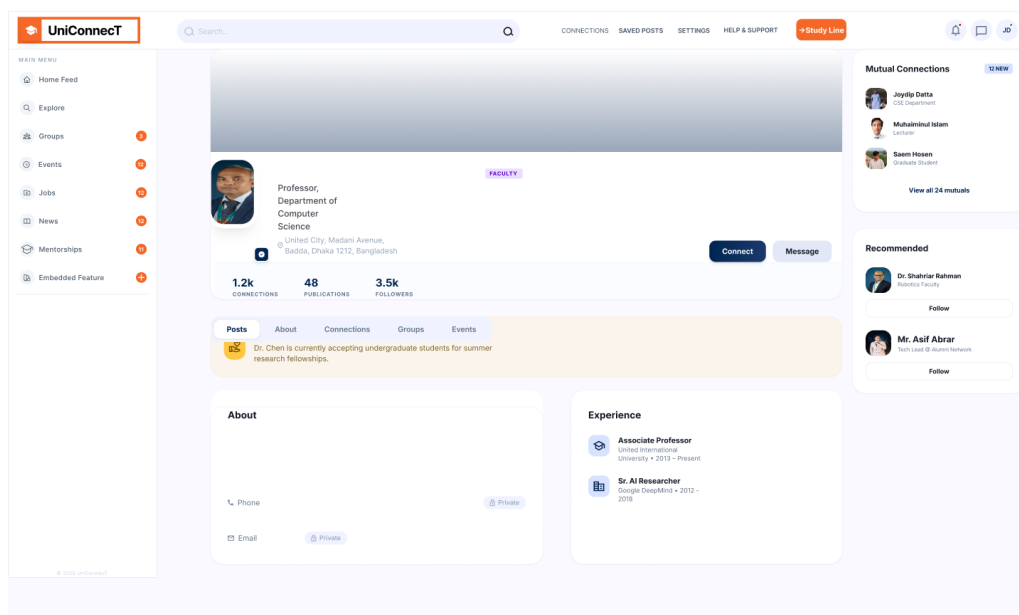


Figure 11: Figma Wireframe — Profile Page. Cover photo, avatar, bio, experience, education, skills, featured items, and profile analytics. Edit controls are visible for the profile owner.

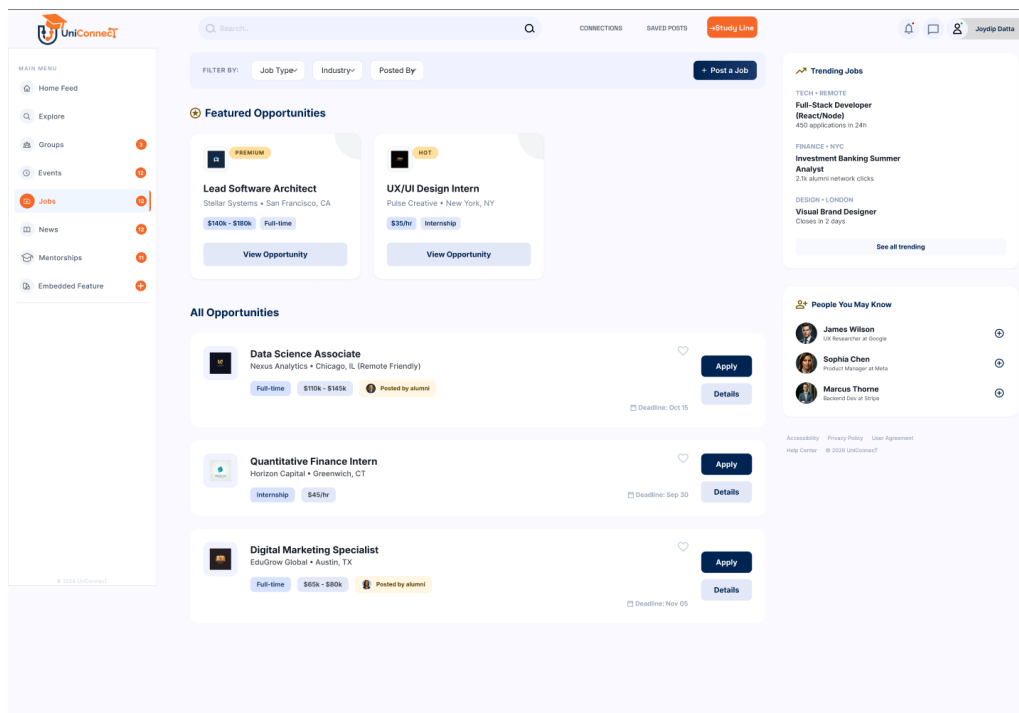


Figure 12: Figma Wireframe — Jobs Page. Filterable and sortable listing of university-internal job and internship postings, with a detail panel for the selected listing.

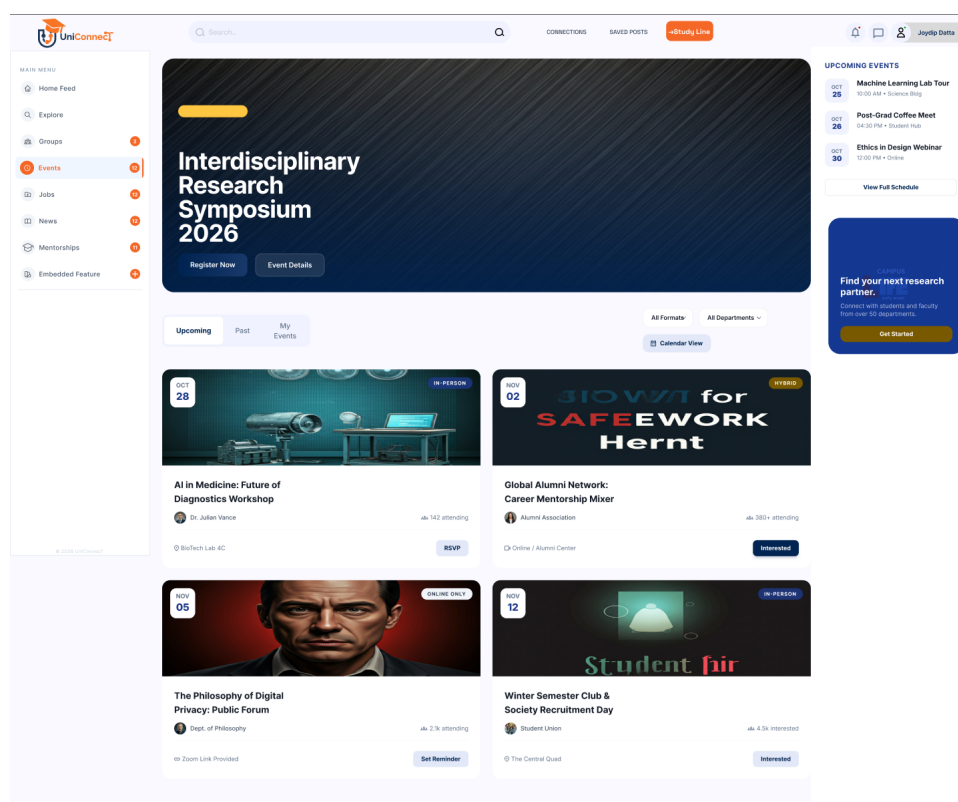


Figure 13: Figma Wireframe — Events Page. Campus event listing with RSVP buttons and a detail view showing event date, location, description, and attendee count.

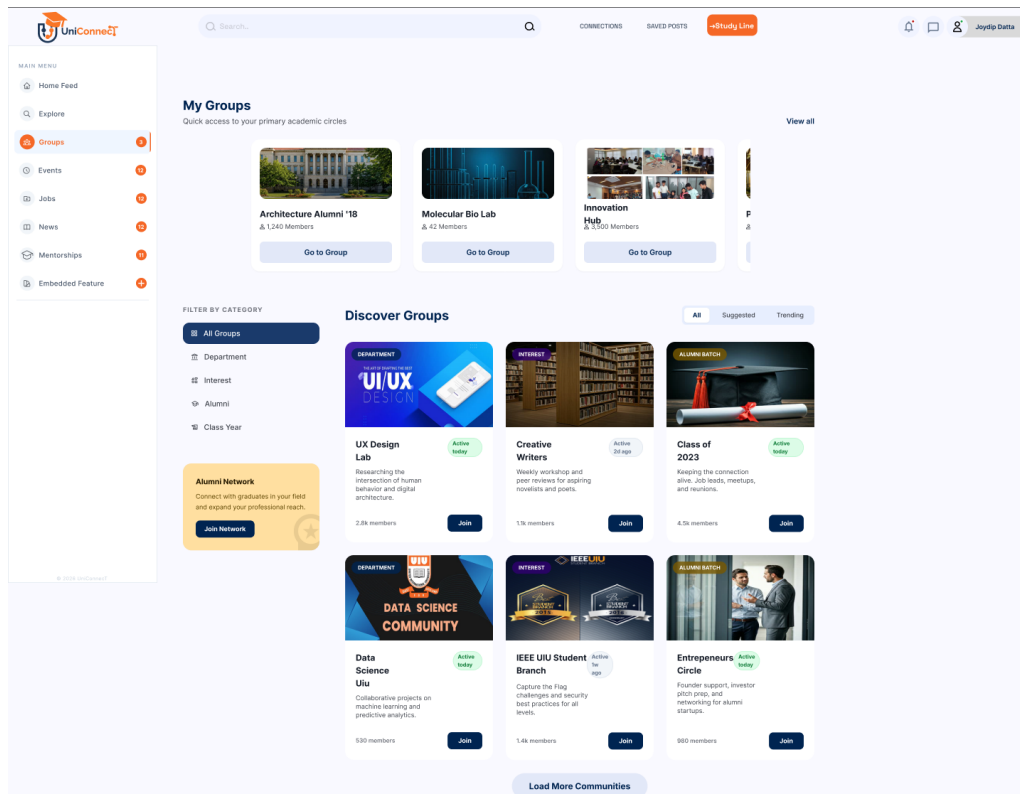


Figure 14: Figma Wireframe — Groups Page. Group directory and a group detail view showing the group feed, members list, file resources, study sessions, pinned posts, and group rules.

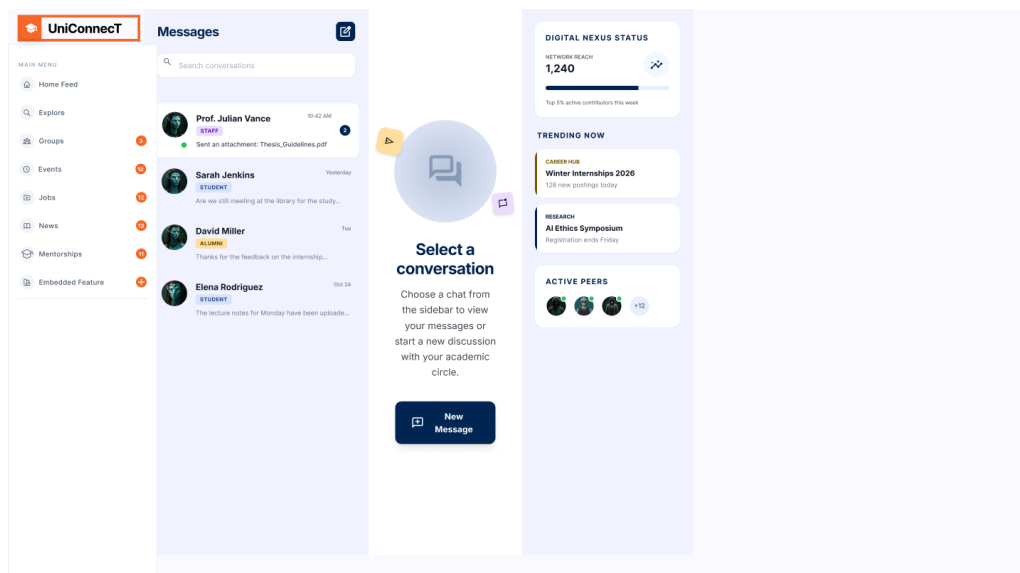


Figure 15: Figma Wireframe — Messages Page. Real-time messaging interface with a conversation list on the left and an active chat panel on the right, supporting direct messages, group chats, and mentorship threads.

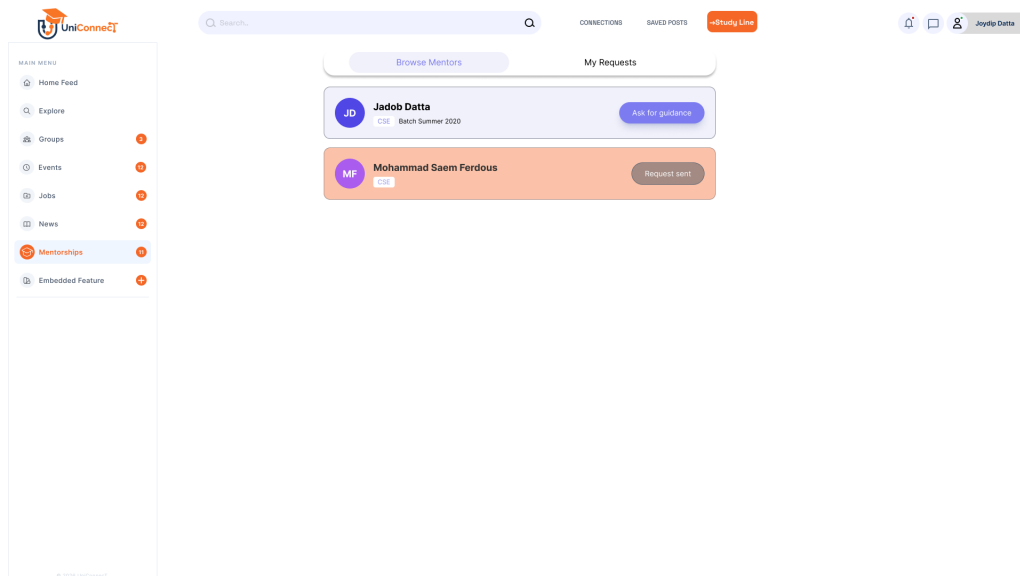


Figure 16: Figma Wireframe — Mentorship Page. Students browse alumni mentor cards and submit requests; alumni manage active mentees, session records, and mentee capacity.

13.3 Campus Utilities and Administration

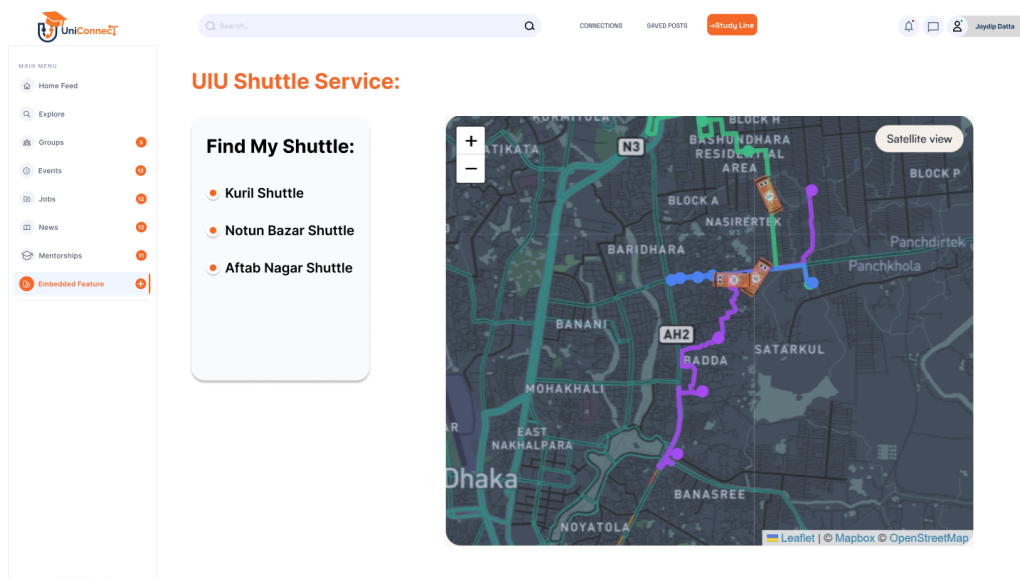


Figure 17: Figma Wireframe — Shuttle Tracker. The UIU Shuttle Service screen lists available shuttle routes (Kuril Shuttle, Notun Bazar Shuttle, Aftab Nagar Shuttle) on the left panel, with a live interactive map of the Dhaka route network on the right. Students can track real-time shuttle positions directly from this view.

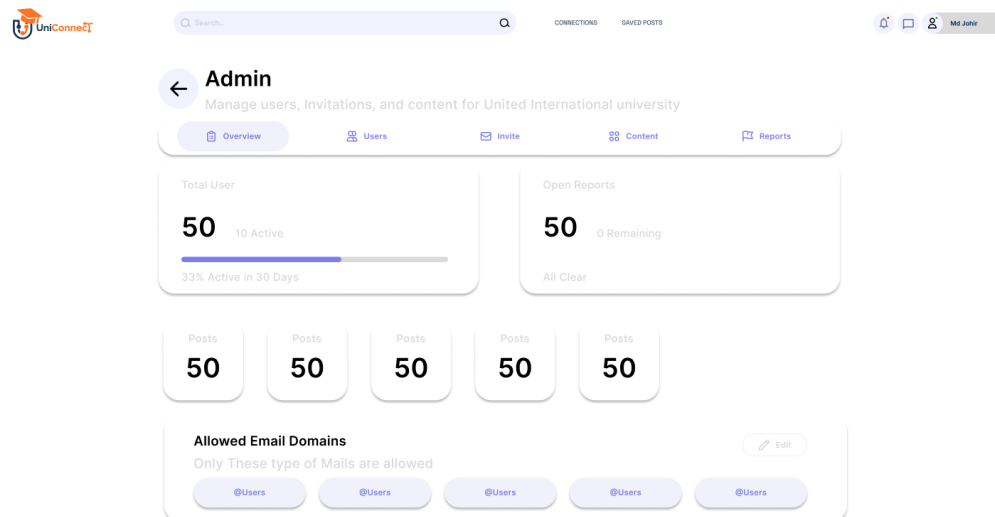


Figure 18: Figma Wireframe — Admin Panel. Role-restricted management interface covering user management, invitation management, content report resolution, platform statistics, and university configuration.

13.4 System Architecture Diagrams

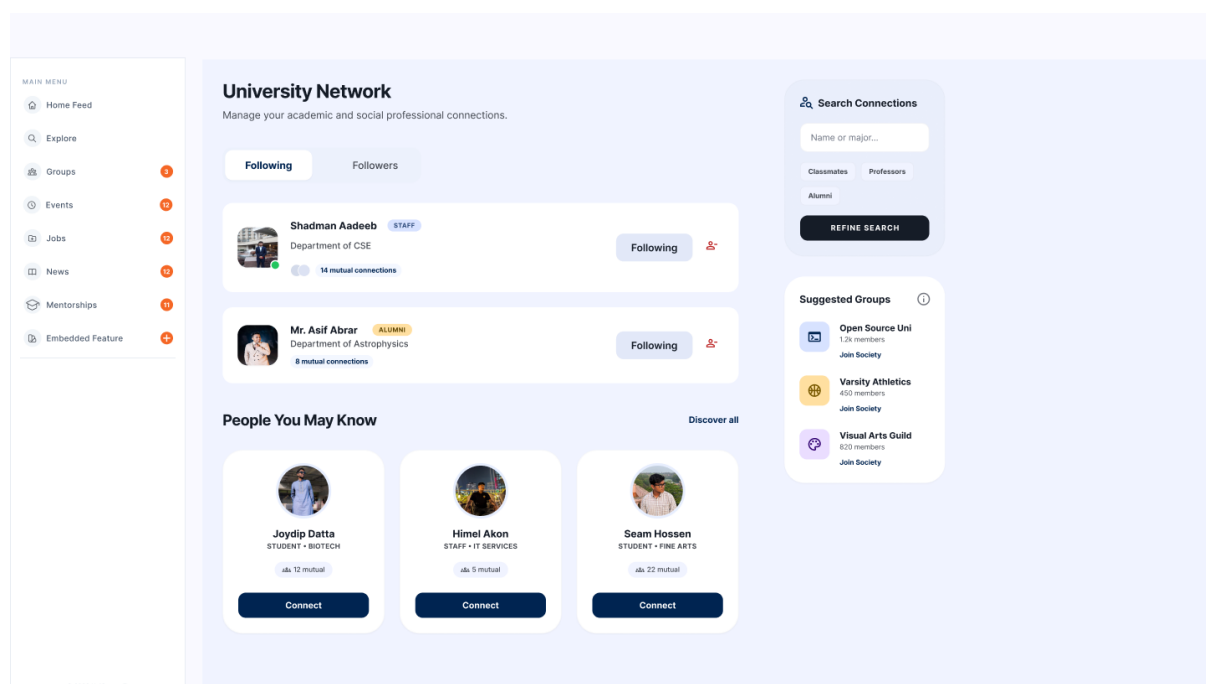


Figure 19: Figma Wireframe — University Network (Connections) Page. Displays the user’s Following and Followers tabs, a “People You May Know” section with mutual-connection counts and Connect actions, and a right-side panel for searching connections by name or major and discovering suggested groups.

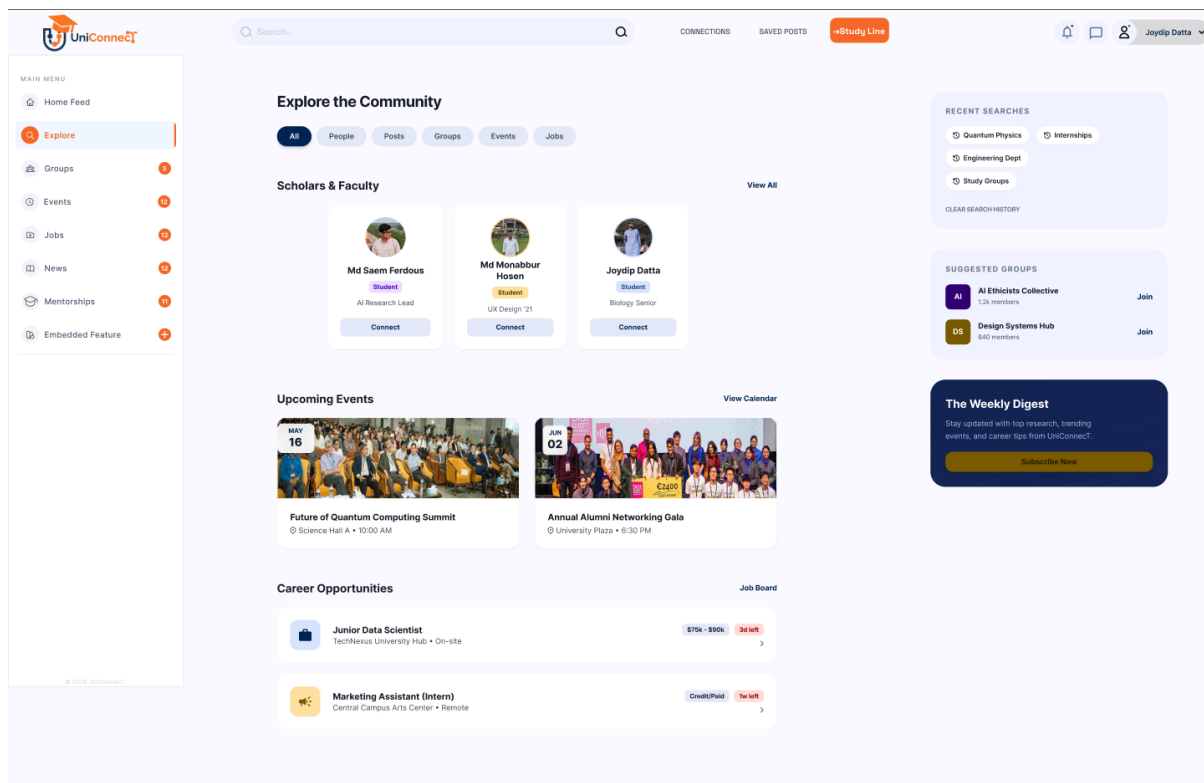


Figure 20: Figma Wireframe — Explore Page. Surfaces community-wide discovery content: a Scholars & Faculty spotlight with connection actions, Upcoming Events with dates and venues, and Career Opportunities pulled from the job board. Filter tabs (All, People, Posts, Groups, Events, Jobs) and a right-side panel showing Recent Searches and Suggested Groups complete the discovery experience.

14. Usability / Design Considerations

14.1 Navigation Flow

UniConnecT uses a persistent top navigation bar on all authenticated pages, providing direct access to the five primary sections: Feed, Jobs, Events, Messages, and Notifications. A left-side profile summary panel on wider screens offers quick links to Profile, Groups, Mentorship, Explore, Lost & Found, Shuttle, and Drafts. The navigation set is role-aware: admin-only controls (Admin Panel) appear only for administrator-role users, and the Driver GPS Broadcast View is exclusive to the driver role. The unauthenticated shell (Landing, Login, Register, OTP, Forgot Password) is entirely separate from the authenticated application and shares no navigation elements with it.

14.2 Responsiveness Across Devices

The design targets three primary breakpoints: mobile (360px–767px), tablet (768px–1023px), and desktop (1024px and above). On mobile, the top navigation bar collapses to a bottom tab bar covering the five primary sections, and the three-column feed layout (feed + two sidebars) reflows to a single-column view. Job and event card grids reflow from three columns on desktop to two on tablet and one on mobile. All interactive touch targets are designed at a minimum of 44×44px on mobile to meet accessibility touch-target guidelines.

14.3 Color Consistency and Design System

UniConnecT is designed with two themes defined in a centralized Figma design system:

- **Warm Futuristic Dark (Default):** Deep navy background surfaces, UIU orange as the primary brand and interactive accent, and indigo for secondary interactive states. Warm white text tokens are used on all dark surfaces. Coloured surfaces (e.g., orange notification badges) use the matching light token for text.
- **Warm Neutral Light:** Off-white and warm grey surfaces with the same UIU orange accent, designed for users who prefer a light interface. Toggled via the user's theme preference in Settings.

No ad-hoc or hardcoded color values appear anywhere in the design. All fills and strokes reference named Figma color styles. Structural borders are 0.5px. All buttons use pill-shaped (fully rounded) border radii exclusively.

14.4 Accessibility Considerations

- **Color Contrast:** All text-on-background color pairings in both themes meet a minimum 4.5:1 contrast ratio (WCAG 2.1 Level AA), verified using the A11y Figma plugin during the design process.
- **Typography:** Body text is set at 14px minimum. Font weight is restricted to 400 (regular) and 500 (medium) throughout. Section headings establish hierarchy through size differences, not color or weight extremes.
- **Interactive States:** All buttons, links, form fields, navigation items, and cards include clearly visible hover, focus, and active states, ensuring keyboard and screen-reader users can navigate effectively.
- **Labels and Tooltips:** All form inputs carry visible text labels. Icon-only controls (e.g., reaction buttons, share, bookmark) include tooltips to communicate purpose without relying solely on icon recognition.
- **Empty and Error States:** Every list, feed, and search result view has a designed empty state with a clear message. Every form includes inline error messages positioned adjacent to the failing field rather than at the top of the page only.

14.5 Key User-Friendly Design Decisions

- **Skeleton Loading States:** The feed, profile, jobs, events, and groups pages use skeleton placeholder cards while content loads, ensuring the interface never presents a blank screen and conveys immediate responsiveness.
- **Role-Aware Navigation:** Navigation surfaces adapt automatically based on the logged-in user's role, so students never encounter admin controls and drivers see only their broadcast interface.
- **Progressive Disclosure:** Secondary and advanced actions (post editing options, advanced group settings, mentorship session notes, privacy controls) are revealed via dropdown menus or modals, keeping primary page views uncluttered and focused.
- **Unified Search:** A single persistent search bar in the top navigation queries all content types simultaneously, with tabbed result views for People, Posts, Jobs, Events, Groups, and News — removing the need for users to navigate to individual sections to search within them.
- **Automatic Mentorship Thread:** When a mentorship request is accepted, a dedicated conversation thread is automatically created and surfaced in the Messages module, eliminating the friction of manually initiating contact after acceptance.

- **Drafts Across Content Types:** A unified Drafts page aggregates all of a user's unpublished content (posts, jobs, news, events) in one place, preventing work from being lost across disconnected creation flows.
- **Consistent Component Language:** Cards, modals, input fields, badges, tabs, and buttons follow a single visual specification across all 24 pages of the design, reducing the visual learning curve for new users and ensuring a coherent, professional-quality interface throughout.

15. Conclusion

UniConnecT is a planned private university social networking platform designed to address the fragmentation and absence of a dedicated digital community space for university students, alumni, faculty, and staff. This Software Requirements Specification documents the system's five intended user roles, 29 functional requirements spanning social networking, professional development, mentorship, campus utilities, real-time communication, and administration, 10 non-functional quality requirements, and the full scope of 24 planned pages and modules — all expressed through the UniConnecT Figma design prototype.

The design goals of UniConnecT center on delivering a unified, role-aware, and accessible platform that connects every member of the university community in a single trusted environment. By grounding the design in a coherent UIU-branded design system (Warm Futuristic Dark and Warm Neutral Light themes), a clear multi-role navigation architecture, and established WCAG 2.1 accessibility standards, the planned interface is intended to provide a modern, student-centric experience that meaningfully bridges academic life, professional development, and everyday campus engagement.

This SRS, prepared by Team Mavericks as part of the System Analysis and Design course at United International University (UIU), serves as the foundational specification for the accompanying Figma prototype and guides all design decisions made throughout the project.